

FULL LAB BUILD REPORT

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Date: 24 June 2026

1. Introduction

Cybersecurity professionals require a safe environment where they can learn, test, and analyze systems without affecting real-world networks or devices. To achieve this, cybersecurity laboratories are created using **virtualization** technologies. These laboratories allow security researchers and ethical hackers to practice various techniques such as **vulnerability assessment** , **web application testing** , **network scanning** , **traffic analysis** , and **penetration testing** in a controlled environment.

In this task, a personal cybersecurity lab was built using **Oracle VirtualBox** and **Kali Linux**. An intentionally vulnerable web application, **OWASP Juice Shop** was deployed inside the lab environment. Various security tools such as **Nmap** , **Burp Suite** , and **Wireshark** were then used to verify the functionality of the lab and analyze network communication.

2. Objective

The primary objective of this task was to create a secure and isolated cybersecurity practice environment that can be used for future **ethical hacking** and **security testing** activities.

The objectives achieved during this task were:

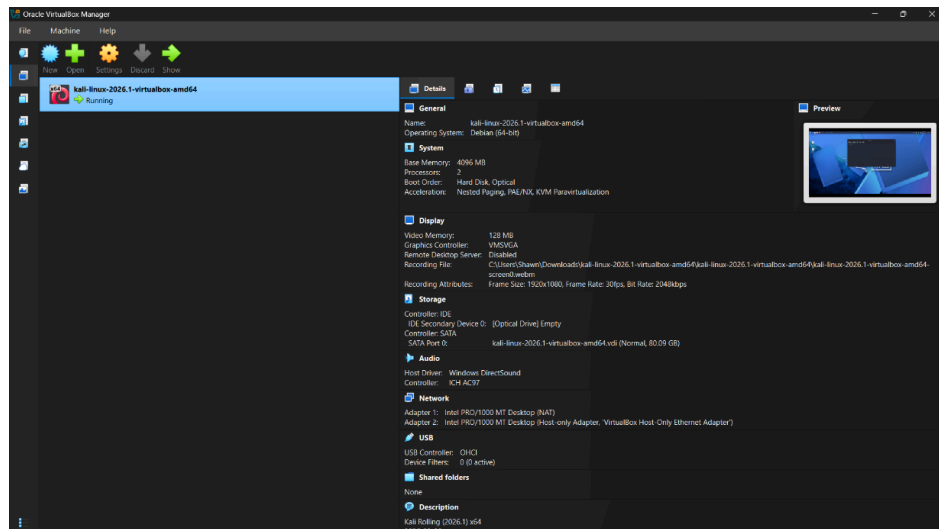
- Installation and configuration of **Oracle VirtualBox**.
- Deployment of **Kali Linux** as the attacker machine.
- Configuration of **NAT** and **Host-Only networking**.
- Installation of **Docker** inside Kali Linux.
- Deployment of **OWASP Juice Shop** using Docker.
- Verification of network communication using **Ping** and **Nmap**.
- Monitoring of web traffic using **Burp Suite**.
- Analysis of network packets using **Wireshark**.

3. System Specifications

The cybersecurity lab was deployed on the following host machine:

Component	Specification
Host Operating System	Windows
Processor	<u>Intel Core i5-12600</u>
RAM	<u>16 GB</u>
Virtualization	Enabled
Virtualization Software	Oracle VirtualBox
Guest Operating System	Kali Linux 2026.1

The available hardware resources were sufficient to run a virtual machine smoothly while simultaneously operating security tools and web applications.



4. Lab Architecture

The lab environment consisted of 1 host machine and 1 virtual machine.

The host machine was running Windows, while Kali Linux was deployed inside Oracle VirtualBox . OWASP Juice Shop was hosted within the Kali Linux environment using Docker containers.

The network architecture was designed as follows:

- **NAT Adapter:** Provided internet access for updates and package installation.
- **Host-Only Adapter:** Provided isolated communication within the cybersecurity lab.

This design ensured that security testing activities remained isolated from external networks while still allowing internet connectivity when required.

5. Installation and Configuration of Kali Linux

A prebuilt Kali Linux VirtualBox image was downloaded and imported into Oracle VirtualBox. The virtual machine was configured with sufficient memory and processing resources to ensure stable performance.

The following resources were allocated:

- 4 GB RAM
- 2 CPU Cores
- NAT Network Adapter
- Host-Only Network Adapter

After importing the virtual machine, Kali Linux was successfully booted and verified.



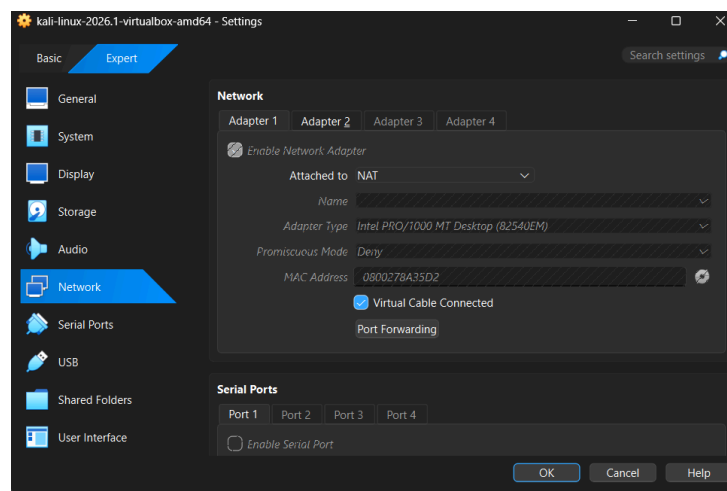
6. Network Configuration

Proper network segmentation is an important requirement in cybersecurity labs because it prevents accidental interaction with external systems during testing.

Two virtual network adapters were configured.

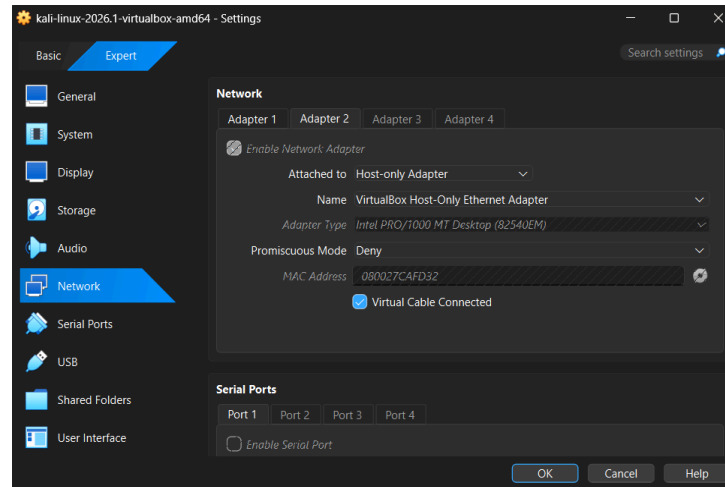
Adapter 1 – NAT

The NAT adapter was used to provide internet access to Kali Linux. This allowed the system to download updates, install packages, and retrieve Docker images.



Adapter 2 – Host-Only Adapter

The Host-Only adapter created an isolated network between the host machine and the Kali Linux virtual machine. This ensured that cybersecurity testing remained within the lab environment.



7. Verification of IP Configuration

After configuring the network adapters, the IP configuration was verified using the following command:

`ip a`

The output confirmed the presence of two active interfaces:

- NAT Interface (Internet Connectivity)
- Host-Only Interface (Lab Communication)

The successful assignment of IP addresses indicated that the network configuration had been completed correctly.

```
kali@kali: ~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 84049sec preferred_lft 84049sec  
    inet6 fd17:025c:1037:2:4b0e:990b:d2fe:2d8b/64 scope global dynamic noprefixroute  
        valid_lft 86091sec preferred_lft 14091sec  
    inet6 fe80::63a4:e12:8c79:70ae/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:ca:fd:32 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1  
        valid_lft 349sec preferred_lft 349sec  
    inet6 fe80::6c07:3c68:44a8:d821/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
kali@kali: ~$
```

8. System Update and Package Verification

Before installing additional tools, the Kali Linux operating system was updated to ensure that all packages were current and security patches were applied.

Commands Used

`sudo apt update`

`sudo apt upgrade -y`

Updating the system helps improve stability, compatibility, and security before deploying new software components.

```
kali@kali: ~  
$ sudo apt update  
[sudo] password for kali:  
The following packages were automatically installed and are no longer required:  
  enchant-2  libio-pty-perl  libpoppler147  libraw23t64  libsodium23  libxmlsec1  openjdk-21-jre-headless  python3-pysnmp4  
  libavc1394-0  libipc-run-perl  libpostal-data  libsql2-classic  libtsan2  libxmlsec1-openssl  postgresql-common-dev  python3-tz  
  libgda13b  libmbedcrypto16  libpostall  libsimdutf31  libunibreak6  openjdk-21-jre  python3-pluginbase  
Use 'sudo apt autoremove' to remove them.  
  
Not upgrading:  
  bulk_extractor  libgtkmm-4.0-0  libqt6gui6  libqt6qmlworkerscript6  libqt6shellintegration6  pyqt6-dev-tools  qt6-wayland  
  gstreamer1.0-plugins-bad  liblayershellqtinterface6  libqt6multimedia6  libqt6quick6  libqt6xml6  python-tables-data  qt6ct  
  layer-shell-qt  libmagiccore-7.q16-10-extra  libqt6network6  libqt6qml6  libqt6termwidget6-2  python3-pyqt6  qterminal  
  libadwaita-1-0  libomxruntime-providers  libqt6opengl6  libqt6sql6-sqlite  libqt6termwidget6-2  python3-tables  qtermwidget-data  
  libfalconsecurityqt64  libpipewire-0.3-0t64  libqt6openglwidget6  libqt6svg6  libspa-0.2-bluetooth  python3-tables-lib  tshark  
  libgpc29t64  libpipewire-0.3-modules  libqt6printsupport6  libqt6test6  libspa-0.2-modules  qt6-base-dev-tools  wireshark  
  libgstreamer-plugins-bad1.0-0  libqt6core8compato  libqt6qml6  libqt6waylandclient6  libspa-0.2-modules  qt6-base-dev-tools  wireshark  
  libgtk-4-1  libqt6core6t64  libqt6qmlmeta6  libqt6waylandcompositor6  pipewire-bin  qt6-gtk-platformtheme  wireshark-common  
  libgtk-4-bin  libqt6dbus6  libqt6qmlmodels6  libqt6widget6  pipewire-pulse  qt6-gpa-plugins  qt6-svg-plugins  
  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 61  
kali@kali: ~$
```

9. Docker Installation

Docker was installed to simplify the deployment of OWASP Juice Shop.

Docker allows applications to run inside containers, making deployment easier and more consistent across different systems.

Commands Used

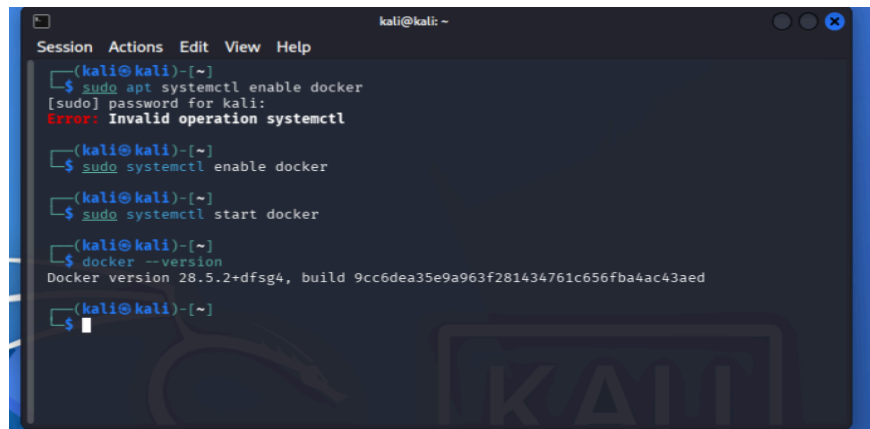
```
sudo apt install docker.io -y
sudo systemctl enable docker
sudo systemctl start docker
docker --version
```

The installation was verified successfully by checking the Docker version.

10. Deployment of OWASP Juice Shop

OWASP Juice Shop is a deliberately vulnerable web application designed for security training and awareness. It contains numerous security vulnerabilities that can be used for ethical hacking practice.

The application was deployed using Docker

A screenshot of a terminal window titled 'kali@kali: ~'. The terminal shows the following commands and output:

```
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo apt systemctl enable docker
[sudo] password for kali:
Error: Invalid operation systemctl
(kali@kali)-[~]
$ sudo systemctl enable docker
(kali@kali)-[~]
$ sudo systemctl start docker
(kali@kali)-[~]
$ docker --version
Docker version 28.5.2+dfsg4, build 9cc6dea35e9a963f281434761c656fba4ac43aed
(kali@kali)-[~]
$
```

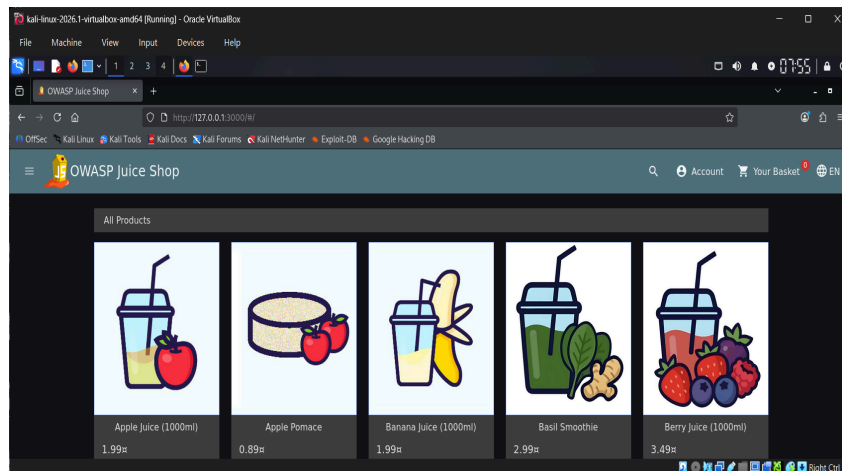
Commands Used

```
sudo docker pull bkimminich/juice-shop
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
```


After deployment, the application was made available on port 3000.

The successful deployment of the Docker container confirmed that the target application was operational.

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ sudo docker pull bkimminich/juice-shop
[sudo] password for kali:
Using default tag: latest
latest: Pulling from bkimminich/juice-shop
Digest: sha256:3c200ce5a59e985c83b4e0bbf78cb7095559c246db6b720f157c97a14bfea75b
Status: Image is up to date for bkimminich/juice-shop:latest
docker.io/bkimminich/juice-shop:latest
(kali@kali)-[~/Desktop]
$ sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
012cefb5c27cb8821444692f906b80ca90bc7581d40f98affd505f0ac72aed
(kali@kali)-[~/Desktop]
$ docker ps
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.51/containers/json": dial unix /var/run/docker.sock: connect: permission denied
(kali@kali)-[~/Desktop]
$ sudo docker ps
CONTAINER ID        IMAGE               COMMAND                  CREATED            STATUS              PORTS
012cefb5c27cb8821444692f906b80ca90bc7581d40f98affd505f0ac72aed  bkimminich/juice-shop  "/nodejs/bin/node /j..."  3 minutes ago     Up 3 minutes        0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
(kali@kali)-[~/Desktop]
$
```



11. Connectivity Testing

Connectivity testing was performed to ensure that communication within the lab network was functioning correctly.

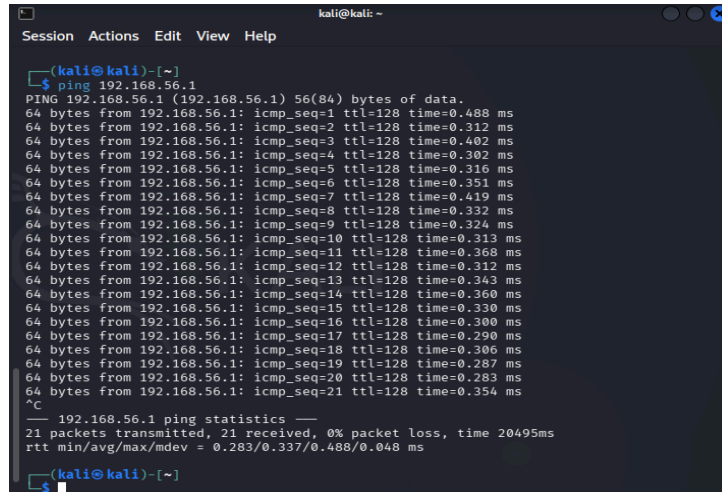
Ping Validation

The Ping utility was used to verify communication between systems.

Command Used

ping 192.168.56.1

The successful responses confirmed that the network interface was reachable.

A terminal window titled 'kali@kali: ~' with a menu bar (Session, Actions, Edit, View, Help). The user enters 'ping 192.168.56.1'. The output shows 21 successful ping responses from 192.168.56.1 to 192.168.56.1, each with 64 bytes of data, TTL=128, and varying times between 0.283ms and 0.488ms. A summary line at the bottom states: '21 packets transmitted, 21 received, 0% packet loss, time 20495ms rtt min/avg/max/mdev = 0.283/0.337/0.488/0.048 ms'.

```
kali@kali: ~  
$ ping 192.168.56.1  
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data:  
64 bytes from 192.168.56.1: icmp_seq=1 ttl=128 time=0.488 ms  
64 bytes from 192.168.56.1: icmp_seq=2 ttl=128 time=0.312 ms  
64 bytes from 192.168.56.1: icmp_seq=3 ttl=128 time=0.402 ms  
64 bytes from 192.168.56.1: icmp_seq=4 ttl=128 time=0.302 ms  
64 bytes from 192.168.56.1: icmp_seq=5 ttl=128 time=0.316 ms  
64 bytes from 192.168.56.1: icmp_seq=6 ttl=128 time=0.351 ms  
64 bytes from 192.168.56.1: icmp_seq=7 ttl=128 time=0.419 ms  
64 bytes from 192.168.56.1: icmp_seq=8 ttl=128 time=0.332 ms  
64 bytes from 192.168.56.1: icmp_seq=9 ttl=128 time=0.324 ms  
64 bytes from 192.168.56.1: icmp_seq=10 ttl=128 time=0.313 ms  
64 bytes from 192.168.56.1: icmp_seq=11 ttl=128 time=0.368 ms  
64 bytes from 192.168.56.1: icmp_seq=12 ttl=128 time=0.312 ms  
64 bytes from 192.168.56.1: icmp_seq=13 ttl=128 time=0.343 ms  
64 bytes from 192.168.56.1: icmp_seq=14 ttl=128 time=0.360 ms  
64 bytes from 192.168.56.1: icmp_seq=15 ttl=128 time=0.330 ms  
64 bytes from 192.168.56.1: icmp_seq=16 ttl=128 time=0.300 ms  
64 bytes from 192.168.56.1: icmp_seq=17 ttl=128 time=0.290 ms  
64 bytes from 192.168.56.1: icmp_seq=18 ttl=128 time=0.306 ms  
64 bytes from 192.168.56.1: icmp_seq=19 ttl=128 time=0.287 ms  
64 bytes from 192.168.56.1: icmp_seq=20 ttl=128 time=0.283 ms  
64 bytes from 192.168.56.1: icmp_seq=21 ttl=128 time=0.354 ms  
^C  
--- 192.168.56.1 ping statistics ---  
21 packets transmitted, 21 received, 0% packet loss, time 20495ms  
rtt min/avg/max/mdev = 0.283/0.337/0.488/0.048 ms  
$
```

12. Network Discovery Using Nmap

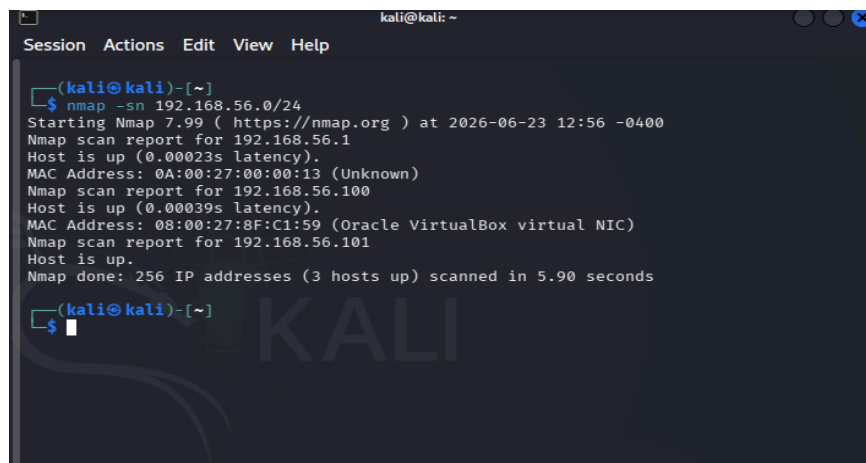
Nmap is one of the most widely used network scanning tools in cybersecurity.

The following command was used to discover active hosts within the Host-Only network.

Command Used

nmap -sn 192.168.56.0/24

The scan successfully identified active devices within the subnet.

A terminal window titled 'kali@kali: ~' with a menu bar (Session, Actions, Edit, View, Help). The user enters 'nmap -sn 192.168.56.0/24'. The output shows the start of an Nmap scan at 2026-06-23 12:56 -0400. It reports three hosts up: 192.168.56.1 (latency 0.00023s), 192.168.56.100 (latency 0.00039s), and 192.168.56.101 (latency 0.00039s). The scan is done in 5.90 seconds, scanning 256 IP addresses.

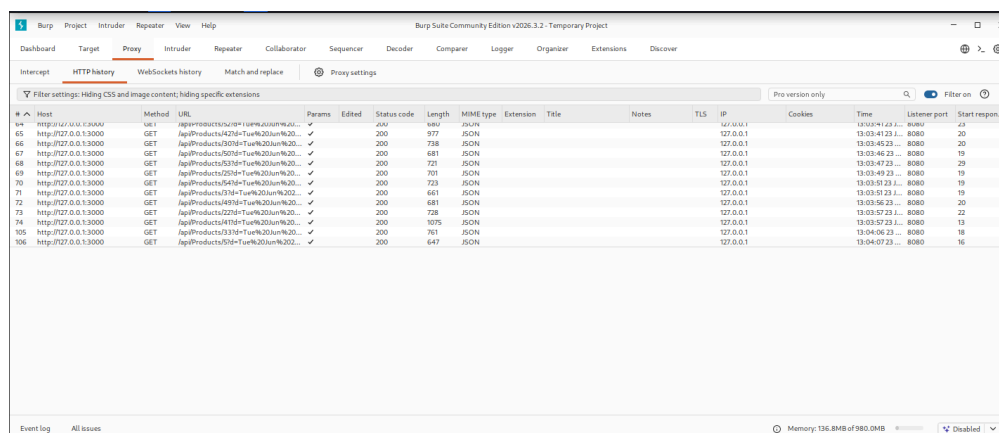
```
kali@kali: ~  
$ nmap -sn 192.168.56.0/24  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-23 12:56 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00023s latency).  
MAC Address: 0A:00:27:00:00:13 (Unknown)  
Nmap scan report for 192.168.56.100  
Host is up (0.00039s latency).  
MAC Address: 08:00:27:8F:C1:59 (Oracle VirtualBox virtual NIC)  
Nmap scan report for 192.168.56.101  
Host is up.  
Nmap done: 256 IP addresses (3 hosts up) scanned in 5.90 seconds  
$
```

13. Web Traffic Analysis Using Burp Suite

Burp Suite Community Edition was used to monitor and inspect HTTP requests generated by OWASP Juice Shop.

The application was accessed through a web browser while Burp Suite captured the requests and responses exchanged between the client and server.

This exercise demonstrated how security professionals inspect web application traffic during vulnerability assessments and penetration testing activities.



#	Host	Method	URL	Params	Edited	Status code	Length	MIME type	Extension	Title	Notes	TLS	IP	Cookies	Time	Listener port	Start response...
64	http://0.0.0.0:3000	GET	api/products/429-Tue%20Jun%20...		✓	200	580	JSON				127.0.0.1	127.0.0.1	13:02:41.23...	8080	20	
65	http://127.0.0.1:3000	GET	api/products/429-Tue%20Jun%20...		✓	200	577	JSON				127.0.0.1	127.0.0.1	13:03:41.23...	8080	20	
66	http://127.0.0.1:3000	GET	api/products/505-Tue%20Jun%20...		✓	200	738	JSON				127.0.0.1	127.0.0.1	13:03:46.23...	8080	20	
67	http://127.0.0.1:3000	GET	api/products/505-Tue%20Jun%20...		✓	200	681	JSON				127.0.0.1	127.0.0.1	13:03:46.23...	8080	19	
68	http://127.0.0.1:3000	GET	api/products/539-Tue%20Jun%20...		✓	200	721	JSON				127.0.0.1	127.0.0.1	13:03:47.23...	8080	20	
69	http://127.0.0.1:3000	GET	api/products/539-Tue%20Jun%20...		✓	200	701	JSON				127.0.0.1	127.0.0.1	13:03:49.23...	8080	19	
70	http://127.0.0.1:3000	GET	api/products/543-Tue%20Jun%20...		✓	200	723	JSON				127.0.0.1	127.0.0.1	13:03:51.23...	8080	19	
71	http://127.0.0.1:3000	GET	api/products/539-Tue%20Jun%20...		✓	200	681	JSON				127.0.0.1	127.0.0.1	13:03:51.23...	8080	19	
72	http://127.0.0.1:3000	GET	api/products/493-Tue%20Jun%20...		✓	200	681	JSON				127.0.0.1	127.0.0.1	13:03:56.23...	8080	20	
73	http://127.0.0.1:3000	GET	api/products/229-Tue%20Jun%20...		✓	200	728	JSON				127.0.0.1	127.0.0.1	13:03:57.23...	8080	22	
74	http://127.0.0.1:3000	GET	api/products/419-Tue%20Jun%20...		✓	200	505	JSON				127.0.0.1	127.0.0.1	13:03:57.23...	8080	13	
105	http://127.0.0.1:3000	GET	api/products/339-Tue%20Jun%20...		✓	200	761	JSON				127.0.0.1	127.0.0.1	13:04:06.23...	8080	18	
106	http://127.0.0.1:3000	GET	api/products/505-Tue%20Jun%20...		✓	200	647	JSON				127.0.0.1	127.0.0.1	13:04:07.23...	8080	16	

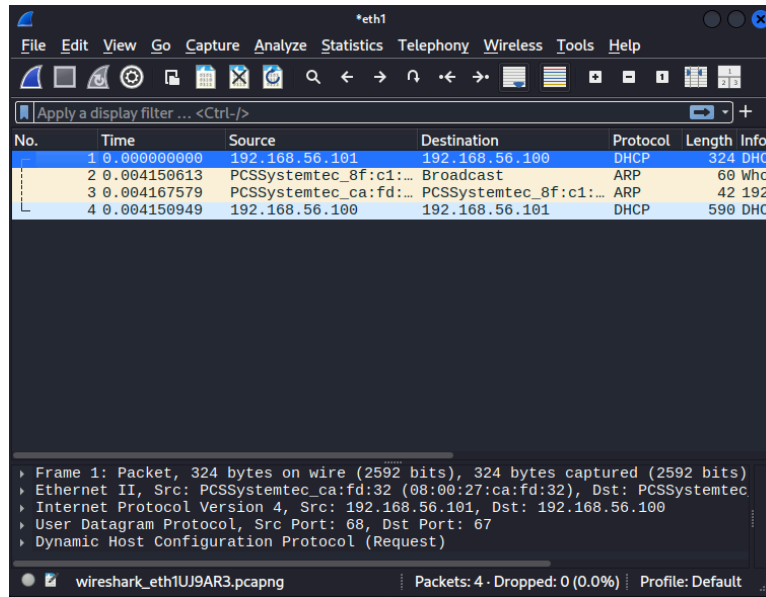
14. Network Traffic Analysis Using Wireshark

Wireshark was used to capture and analyze packets transmitted within the cybersecurity lab.

Traffic was generated by interacting with OWASP Juice Shop while Wireshark monitored the network interface.

The captured packets provided visibility into the communication taking place between the browser and the web application.

This activity demonstrated the importance of packet analysis in incident response, troubleshooting, and network security investigations.



15. Challenges Faced and Solutions

During the implementation of this lab, several challenges were encountered:

- Configuring dual network adapters correctly.
- Understanding the difference between NAT and Host-Only networking.
- Managing Docker container permissions.
- Verifying communication between different interfaces.

These issues were resolved through troubleshooting, configuration verification, and practical testing.

The experience improved my understanding of virtualization, networking, Docker containers, and cybersecurity tools.

16. Conclusion

The cybersecurity laboratory was successfully designed and deployed using **Oracle VirtualBox, Kali Linux, Docker, and OWASP Juice Shop**.

The lab environment provides a safe platform for learning and practicing cybersecurity concepts without impacting real-world systems. Network

connectivity was verified using Ping and Nmap, web traffic was analyzed using Burp Suite, and packet-level communication was examined using Wireshark.

This task significantly improved my understanding of **virtualization, network isolation, web application deployment, security testing tools, and practical cybersecurity workflows**. The completed lab environment can now be used for future penetration testing exercises, vulnerability assessments, and hands-on cybersecurity learning.